

Feature extraction formulas

Perfect.

Now we go into **mathematically exact feature extraction specification**.

This will be written so your AI agent can directly implement it without interpretation.

No vague wording. Pure formulas + algorithm steps.

PIECE 5 — COMPLETE FEATURE EXTRACTION FORMULAS

Input Definition

Given:

- Sampling rate: **16,000 Hz**
- Clip duration: **2 seconds**
- Total samples:

[
 $N = 32,000$
]

Let the audio signal be:

[
 $x[n], \quad n = 0, 1, 2, \dots, 31999$
]

Assume values are normalized to:

[
 $x[n] \in [-1, 1]$
]

Step 1 — Frame Division

Total frames: **20**

Frame size:

[
 $F = 1600 \text{ \text{samples}}$
]

Frame index:

[
 $k = 0, 1, \dots, 19$
]

Frame samples:

$$[\\begin{aligned} x_k[m] &= x[k \\cdot F + m] \\\\ \\end{aligned}$$

Where:

$$[\\begin{aligned} m &= 0, 1, \\dots, 1599 \\\\ \\end{aligned}$$

Feature 1 — Total Energy

Definition:

$$[\\begin{aligned} E_{\\text{total}} &= \\sum_{n=0}^{N-1} x[n]^2 \\\\ \\end{aligned}$$

Optional normalization:

$$[\\begin{aligned} E_{\\text{total}} &= \\frac{1}{N} \\sum x[n]^2 \\\\ \\end{aligned}$$

(Recommended to divide by N for scale stability.)

Feature 2 — Mean Amplitude

Definition:

$$[\\begin{aligned} \\mu &= \\frac{1}{N} \\sum_{n=0}^{N-1} x[n] \\\\ \\end{aligned}$$

This captures DC offset (should be near zero).

Feature 3 — Signal Variance

Definition:

$$[\\begin{aligned} \\sigma^2 &= \\frac{1}{N} \\sum_{n=0}^{N-1} (x[n] - \\mu)^2 \\\\ \\end{aligned}$$

Measures amplitude spread.

Feature 4 — Zero Crossing Rate (ZCR)

Definition:

Count sign changes:

$$ZCR = \frac{1}{N-1} \sum_{n=1}^{N-1} \mathbf{1}_{\{x[n] \cdot x[n-1] < 0\}}$$

Where:

$$\mathbf{1}_{\{\text{condition}\}} = \begin{cases} 1 & \text{if true} \\ 0 & \text{if false} \end{cases}$$

ZCR measures signal frequency characteristics.

Clap → high ZCR

Silence → near zero

Frame-Based Energy

For each frame (k):

$$E_k = \frac{1}{F} \sum_{m=0}^{F-1} x_k[m]^2$$

This produces:

$$E_0, E_1, \dots, E_{19}$$

Feature 5 — Mean Frame Energy

$$\mu_E = \frac{1}{20} \sum_{k=0}^{19} E_k$$

This captures average temporal energy.

Feature 6 — Variance of Frame Energy

$$\left[\begin{aligned} \sigma_E^2 &= \frac{1}{20} \sum_{k=0}^{19} (E_k - \mu_E)^2 \end{aligned} \right]$$

This captures how “spiky” the sound is.

Clap → high variance

Noise → low variance

Final Feature Vector

The 6-dimensional feature vector is:

$$\left[\begin{aligned} X &= \\ \begin{bmatrix} E_{\text{total}} \\ \mu \\ \sigma^2 \\ \text{ZCR} \\ \mu_E \\ \sigma_E^2 \end{bmatrix} \end{aligned} \right]$$

Return as:

```
std::vector<double> features(6);
```

Order must remain consistent.

Important Implementation Constraints

1. Use double precision.
2. Always divide by N or F to prevent scale explosion.
3. No log operations here.
4. Do NOT compute normalization here.
5. FeatureExtractor must be stateless.

Computational Complexity

Per sample:

- $O(N)$ for total energy
- $O(N)$ for ZCR

- $O(N)$ for mean & variance
- $O(N)$ for frame energy

Total:

[
 $O(N)$
]

Efficient enough for 200 samples.

Why These Features Work

Clap:

- High total energy
- High frame variance
- Possibly high ZCR

Knock:

- Moderate energy
- Medium frame variance

Noise:

- Steady energy
- Low frame variance

Silence:

- Near zero energy
 - Near zero variance
-

Feature Extraction Algorithm Summary

1. Input raw samples (size 32,000)
2. Compute:
 - Total energy
 - Mean
 - Variance
 - ZCR
3. Divide into 20 frames
4. Compute frame energies

5. Compute:

- Mean frame energy
- Variance frame energy

6. Return 6-element vector